MATHS AA SL · Probability, simulation and decisions

Worked mark scheme · 35 minutes · 28 marks

Award equivalent valid methods and conclusions. Apply the indicated mark allocation to each part. Credit consistent follow-through where the subsequent reasoning remains valid.

1. Two different faces [6]

(a) **[3]**

Find the probability that the score recorded is 3.

M1 · There are 16 equally likely ordered outcomes.

M1 · Count outcomes with both values at most 3 and subtract those with both at most 2.

A1 · $(9-4)/16 = 5/16$.

(b) **[3]**

In 256 independent trials, give the expected number recording 3. A student predicts exactly this many. Comment.

M1 · Multiply the event probability by 256.

A1 · The expectation is 80.

R1 · Actual counts vary between runs. An expectation is a long-run average.

2. A fair simulation from digits [7]

(a) **[2]**

Calculate the probability of two reds in the first two colours under Lin's rule. Compare with a fair game.

M1 · Use independence to square the probability of red.

A1 · Lin gives 0.16, whereas fairness gives $1/9$.

(b) **[3]**

Design a colour rule using the same generator that gives three equally likely colours. Justify it.

M1 · Assign three digits apiece, for example 0–2, 3–5 and 6–8.

A1 · Discard 9 and draw again.

R1 · Conditional on acceptance each group has probability $3/9$.

(c) **[2]**

For your rule using one discarded digit, find the probability that at least one redraw is required while producing the first four colours.

M1 · Use the complement of four immediate acceptances.

A1 · $1 - 0.9^4 = 0.3439$.

3. A second check on a parcel [8]

(a) **[2]**

Find the probability it is flagged.

M1 · Add the two mutually exclusive routes to a flag.

A1 · $0.12(0.85) + 0.88(0.05) = 0.146$.

(b) **[3]**

Given that it is flagged, find the probability it needs relabelling. Explain why the answer differs from 0.85.

M1 · Divide $0.12(0.85)$ by 0.146.

A1 · $51/73 \approx 0.699$.

R1 · The 85% is conditioned on a parcel needing relabelling. The group of flagged parcels also contains false alarms.

(c) **[3]**

The false-flag rate is adjustable. Find the largest false-flag rate for which at least 80% of flagged parcels need relabelling, with other rates fixed.

M1 · Set $0.102/(0.102 + 0.88f) \geq 0.8$.

M1 · Rearrangement gives $0.0255 \geq 0.88f$.

A1 · $f \leq 0.02898$ approximately, so at most 2.90% to three significant figures.

4. Two gates at a venue [7]

(a) **[2]**

Estimate the probability of a long wait for a randomly selected visitor in that interval.

M1 · Pool the counts of long waits and visitors.

A1 · $22/200 = 0.11$.

(b) **[3]**

On another evening 40% use the east gate. Assuming the gate-specific rates stay the same, estimate the number of long waits among 600 visitors.

M1 · Weight 0.15 and 0.05 by 0.4 and 0.6.

A1 · The probability is 0.09.

A1 · Expect 54 visitors.

(c) **[2]**

A manager attributes any change from 11% solely to faster service. Evaluate that conclusion.

R1 · The mix of gates changed.

R1 · With both gate-specific wait rates unchanged the overall estimate still falls by two percentage points.

End of mark scheme · 28 marks